Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

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Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
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Name	Saipranav M	Register No.	3122235001110
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Experiment 2: Binary Classification using Naïve Bayes and K-Nearest Neighbors

1. Aim and Objective

The aim of this experiment is to implement and analyze binary classification models using Naïve Bayes and K-Nearest Neighbors (KNN). The performance of each classifier is evaluated using appropriate metrics, visualizations, and hyperparameter tuning techniques to understand generalization behavior, bias–variance trade-offs, and model efficiency.

2. Dataset Description

The Spambase dataset consists of numerical features extracted from email messages. Each record represents an email and is labeled as either spam or non-spam. The dataset is widely used for evaluating binary classification algorithms in text-based problems.

Dataset Source:

- Kaggle – Spambase Dataset

3. Preprocessing Steps

- Loaded the dataset using the Pandas library
- Separated input features and target labels
- Performed train–test split while maintaining class distribution
- Applied feature scaling using StandardScaler to normalize numeric values
- Verified class balance before model training

4. Implementation Details

- Implemented Gaussian, Multinomial, and Bernoulli Naïve Bayes classifiers
- Built an initial KNN classifier as a baseline model
- Tuned KNN hyperparameters using GridSearchCV and RandomizedSearchCV

- Evaluated KDTree and BallTree algorithms for neighbor search efficiency
- Compared models using accuracy, precision, recall, F1-score, ROC curve, and confusion matrix

5. Visualizations

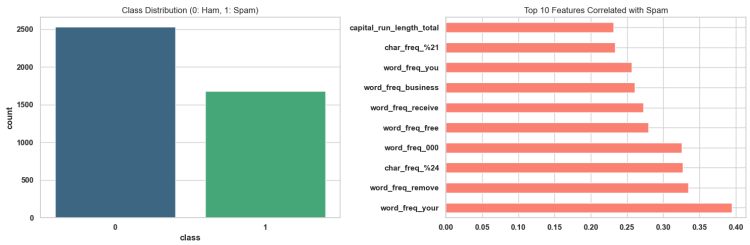

Figure 1: Spam vs Non-Spam Class Distribution

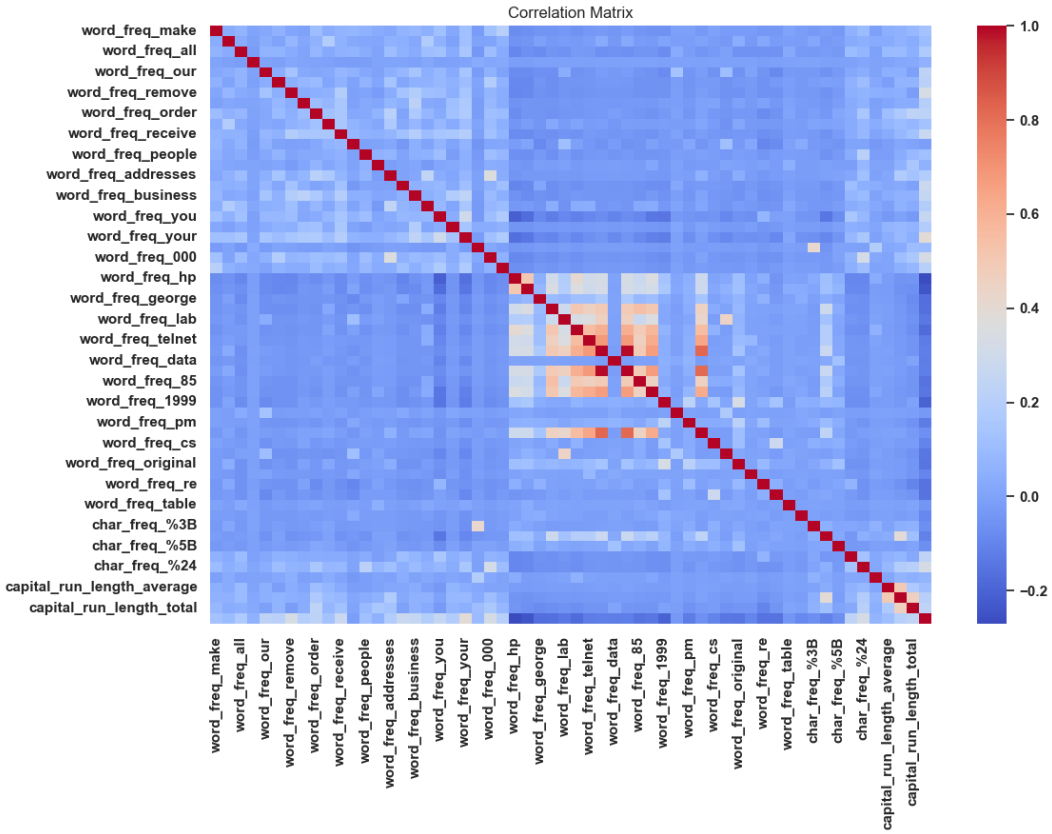

Figure 2: Feature Correlation Matrix

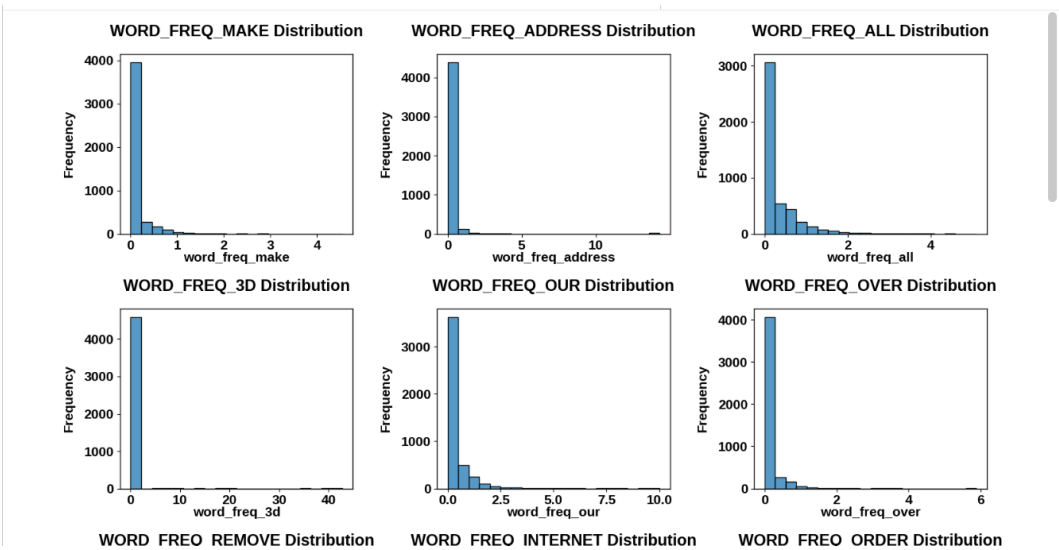

Figure 3: Feature Distribution using Histogram

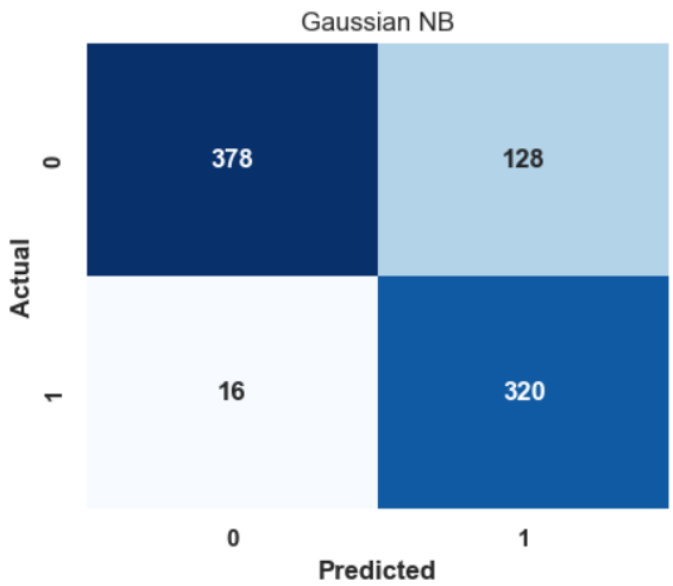

Figure 4: Confusion Matrix – Gaussian Naïve Bayes

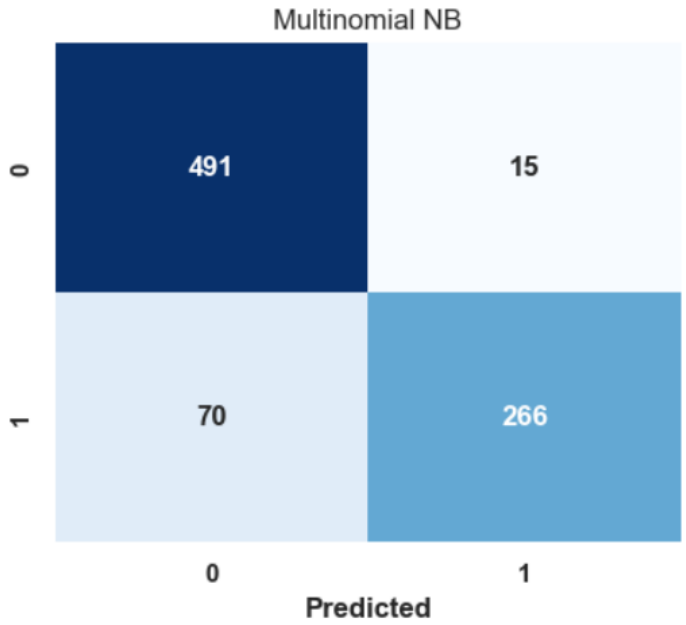

Figure 5: Confusion Matrix – Multinomial Naïve Bayes

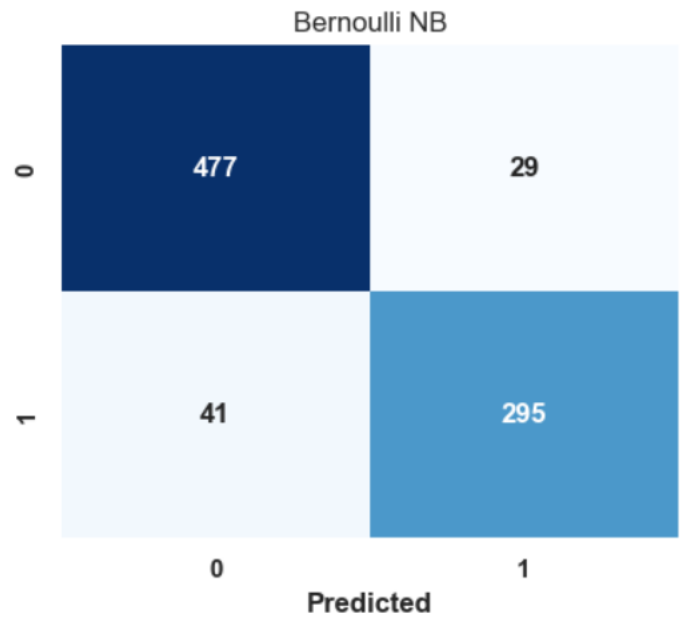

Figure 6: Confusion Matrix – Bernoulli Naïve Bayes

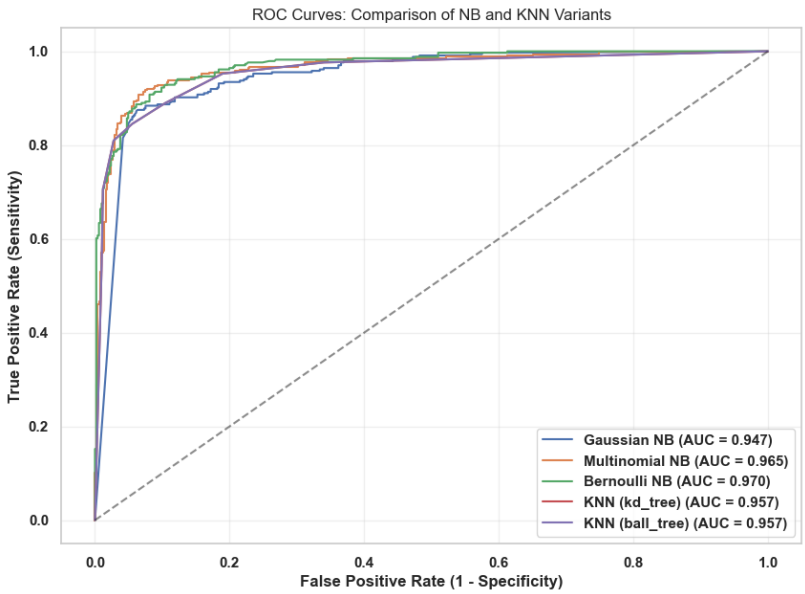

Figure 7: ROC Curve – Naïve Bayes vs KNN

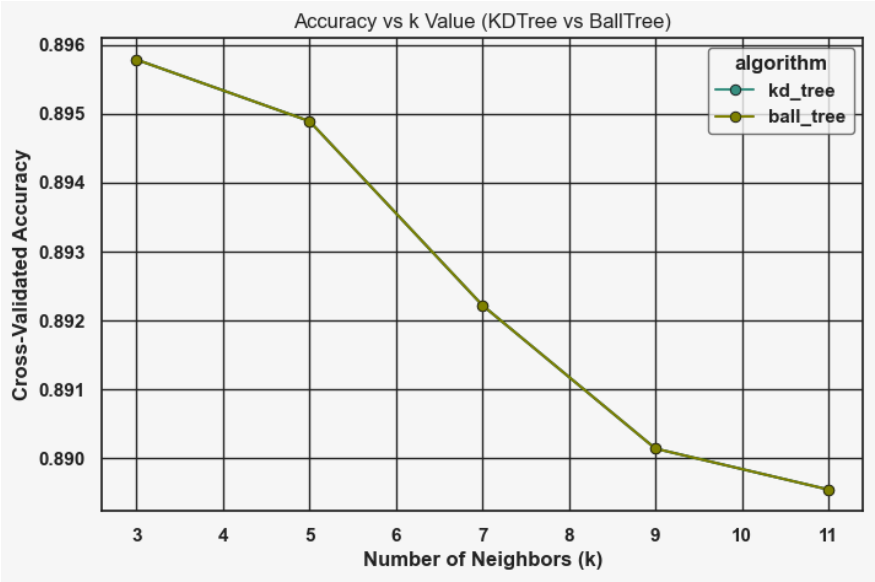

Figure 8: Accuracy vs K Value for KNN (Overlapping since they vary only by computational efficiency.

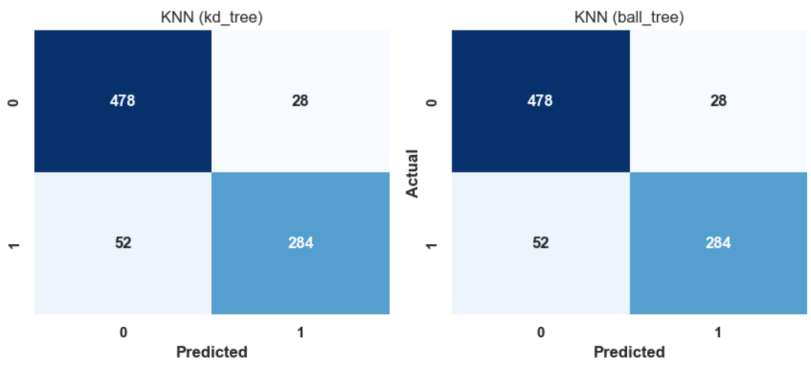

Figure 9: Confusion Matrix – KNN Models

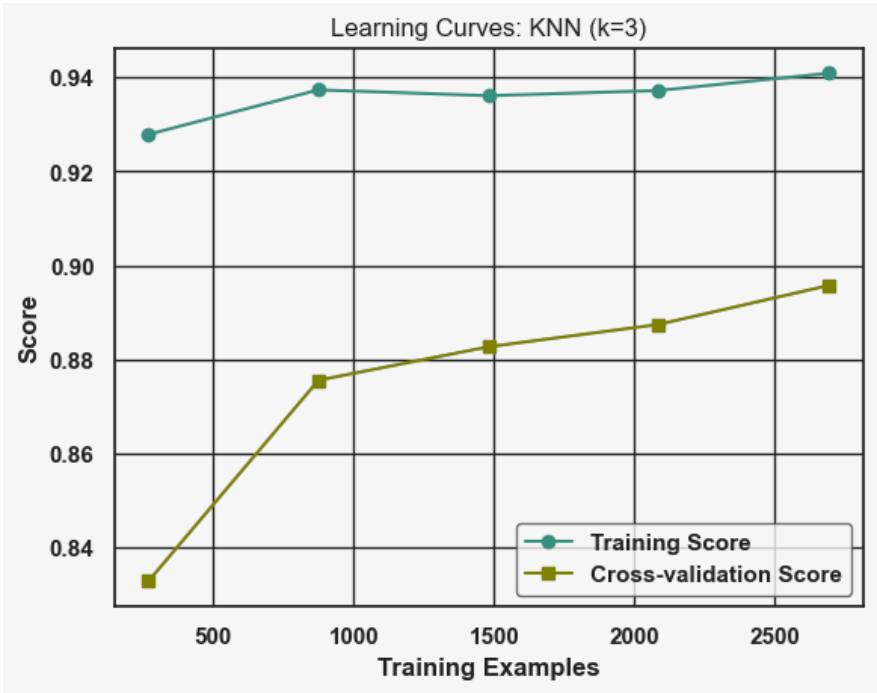

Figure 10: Training vs Validation Accuracy for KNN

6. Performance Tables

Naïve Bayes Performance Metrics

Metric	Gaussian NB	Multinomial NB	Bernoulli NB
Accuracy	0.8289	0.8990	0.9169
Precision	0.7143	0.9466	0.9105
Recall	0.9524	0.7917	0.8780
F1 Score	0.8163	0.8622	0.8939
Specificity	0.7470	0.9704	0.9427
AUC	0.9471	0.9655	0.9697

KNN Hyperparameter Optimization

Search Method	Best Parameters	CV Accuracy
Cross Validation	k = 3, KDTree	0.8958
Cross Validation	k = 3, BallTree	0.8958

Final KNN Performance

Metric	KDTree	BallTree
Optimal k	3	3
Accuracy	0.9097	0.9097
Precision	0.9063	0.9063
Recall	0.8631	0.8631
F1 Score	0.8841	0.8841
Specificity	0.9407	0.9407
Training Time (s)	0.0138	0.0084
Prediction Time (s)	0.0882	0.1027

7. Overfitting and Underfitting Analysis

- When very small values of the parameter k were used in KNN, the model closely followed the training data, resulting in high training accuracy but reduced performance on unseen test samples. This behavior indicated overfitting.
- When very large values of k were selected, the model became overly smooth and failed to capture important local patterns in the data, leading to underfitting.
- An intermediate value of k provided a better balance between learning meaningful patterns and avoiding noise in the dataset.
- Cross-validation and hyperparameter tuning helped identify the optimal k value that produced stable and consistent performance.
- This analysis demonstrates the importance of selecting appropriate model complexity to avoid both overfitting and underfitting.

8. Bias–Variance Analysis

- Naïve Bayes classifiers introduce higher bias due to the assumption of conditional independence between features, which limits the model's flexibility.
- Despite this limitation, Naïve Bayes performs well for high-dimensional text data because of its simplicity and fast computation.
- The K-Nearest Neighbors (KNN) algorithm exhibits higher variance when smaller values of k are used, making the model sensitive to noise in the training data.
- Increasing the value of k reduces variance but increases bias, affecting the model's ability to learn complex patterns.
- Proper hyperparameter tuning enables a balanced bias–variance trade-off, improving the model's generalization capability.

9. Observations and Conclusion

From the experimental results, it was observed that Naïve Bayes classifiers required minimal training time and provided stable baseline performance for spam classification. Among the variants, Bernoulli Naïve Bayes achieved the highest accuracy due to the binary nature of the dataset features.

The optimized KNN classifier produced higher predictive accuracy compared to Naïve Bayes after selecting an appropriate value of k. Both KDTree and BallTree neighbor search strategies resulted in similar classification performance; however, BallTree demonstrated slightly better computational efficiency during prediction.

Overall, the experiment highlights the importance of data preprocessing, exploratory data visualization, and systematic hyperparameter optimization in building reliable machine learning models.

References

- Scikit-learn Documentation – Naïve Bayes
- Scikit-learn Documentation – K-Nearest Neighbors
- Scikit-learn Documentation – Hyperparameter Optimization
- Kaggle – Spambase Dataset